

ANALYSTLAB AFRICA — HEALTHCONNECT EXPERIENCE LAB

Week 4: Initial Analysis Document — Data Analytics Track

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This document establishes the foundation for the Data Analytics track's contribution to the HealthConnect Clinic Experience Lab. It covers a review of the appointment dataset, an initial data quality assessment, the business questions the analysis will pursue, candidate KPIs linked to those questions, and the proposed approach for the analysis phase ahead.

1. Dataset Overview

The HealthConnect_Appointment_Data.csv file contains 5,000 fictional, anonymised appointment records across 18 fields, covering patient demographics, booking and appointment details, appointment history, reminder information, distance to clinic, waiting time, and appointment outcome. Each row represents a single appointment, identified by a unique appointment_id, with patient_id allowing appointments to be linked back to the same patient where relevant.

- No duplicate appointment_id values — the dataset is clean at the record-identity level.
- appointment_outcome splits into three categories: No-Show (2,423 records, 48.5%), Attended (2,314, 46.3%), and Cancelled (263, 5.3%). The 48.5% No-Show proportion is a notable characteristic of this synthetic dataset and should not be assumed to represent real-world clinic attendance patterns.
- appointment_type has four categories: General Consultation (2,086), Follow-up (1,421), Specialist Consultation (900), and Diagnostic Test (593).
- gender includes Female (2,488), Male (2,404), and Prefer not to say (108).
- age ranges from 18 to 80, with a median of 49 and a mean of approximately 48.8.
- booking_lead_days ranges from 0 to 60 days with no negative values, and its values match the difference between booking_date and appointment_date across all 5,000 records.
- previous_no_shows never exceeds previous_appointments for any record, which is logically consistent with the historical count fields.

2. Data Quality Assessment

The initial assessment found no duplicate appointment IDs and no violations of the logical constraints checked during the assessment. Three fields contain missing values, as summarised below.

Field	Missing	Likely Cause	Proposed Handling
reminder_channel	1,366 (27.3%)	Structural, not an error — missing exactly where reminder_sent = "No" (1,366 of 1,366 records). No reminder was sent, so there is no channel to record.	Keep as-is; treat the blank as "No reminder sent" rather than imputing a channel.
distance_to_clinic_km	90 (1.8%)	A small amount of missing data was identified. No specific missingness pattern was investigated at this stage.	Proposed approach: exclude affected rows from distance-specific analysis only, while retaining them for analyses that do not require this field.
waiting_time_minutes	60 (1.2%)	A small amount of missing data was identified. No specific missingness pattern was investigated at this stage.	Proposed approach: exclude affected rows from waiting-time-specific analysis only, while retaining them for analyses that do not require this field.

All other fields (appointment_id, patient_id, gender, age, age_group, appointment_type, booking_date, appointment_date, appointment_day, appointment_time, booking_lead_days, previous_appointments, previous_no_shows, reminder_sent, appointment_outcome) have zero missing values.

A further check will be needed once deeper analysis begins: confirming that appointment_day and age_group are correctly derived from appointment_date and age respectively across the full dataset, since these are derived fields. At this stage, these fields have not been fully validated against their source variables.

3. Relevant Business Questions

- Q1. What factors are most associated with appointment no-shows, including age, distance to the clinic, appointment type, booking lead time, and previous no-show history?
- Q2. Does sending a reminder relate to attendance, and does the reminder channel used make a difference?
- Q3. Does booking lead time (how far in advance an appointment is booked) relate to the likelihood of a no-show?
- Q4. Are patients with a history of previous no-shows more likely to miss future appointments, or is no-show behaviour largely one-off?
- Q5. Does distance to the clinic relate to attendance, and could this inform outreach, reminders, or scheduling decisions?

4. Potential KPIs

The KPIs below are identified and justified at a conceptual level only, in line with the Week 4 scope. None have been calculated, analysed, or visualised yet. Calculation and visualisation are planned for the later analysis phase.

KPI	Linked Question	Relevance / Justification
Overall No-Show Rate	Q1	Establishes the baseline attendance problem the clinic is facing. Other KPIs can be interpreted relative to this figure.
No-Show Rate by Reminder Status / Channel	Q2	Tests whether reminders are associated with attendance and allows comparison across SMS, WhatsApp, and Email.
No-Show Rate by Booking Lead Time Band	Q3	Grouping <code>booking_lead_days</code> into bands can help assess whether the time between booking and appointment is associated with no-show behaviour.
No-Show Rate Among Patients With Previous No-Shows	Q4	Compares the no-show rate among patients with <code>previous_no_shows > 0</code> against those with none, to assess whether prior history is associated with recurrence.
No-Show Rate by Distance Band	Q5	Grouping <code>distance_to_clinic_km</code> into bands can help assess whether travel distance is associated with attendance.

At the calculation stage, the fourth KPI will specifically represent the no-show rate among patients with at least one previous no-show, compared with the no-show rate among patients without a previous no-show history. It is not intended to measure how many times individual patients repeatedly no-show.

5. Initial Analysis Approach

- Load and validate the dataset in Python (Pandas), confirming the data quality findings above and checking the derived fields (`age_group` and `appointment_day`) for consistency.
- Handle missing values according to the proposed strategy in Section 2 — no `reminder_channel` imputation, and row exclusion scoped only to analyses that require `distance_to_clinic_km` or `waiting_time_minutes`.
- Calculate the five candidate KPIs and examine each against `appointment_outcome` using cross-tabulations and grouped summary statistics, while avoiding assumptions of causation.
- Where patterns emerge, cross-check them against more than one variable (for example, distance combined with appointment type) before drawing conclusions, to reduce the risk of interpreting a single-variable pattern as a meaningful driver.
- Build an initial Power BI dashboard in the following phase to visualise the confirmed KPIs and support the clinic's decision-making, once the underlying analysis has been validated in Python.
- Treat findings as descriptive and correlational at this stage; the available data and planned analysis do not by themselves establish causal reasons for missed appointments.

6. Assumptions, Limitations, Risks and Dependencies

- The dataset is fictional and synthetic. The 48.5% No-Show proportion is a characteristic of this dataset and should not be interpreted as representative of real clinic attendance patterns.
- `previous_appointments` and `previous_no_shows` are described as synthetic historical counts. They may not represent a complete or coherent real appointment history for each `patient_id`, so patient-level trend analysis should be treated cautiously.
- `distance_to_clinic_km` and `waiting_time_minutes` each contain a small amount of missing data (under 2%). The proposed approach is to exclude affected rows only from analyses that require those fields, while retaining them for other analyses.
- This document reflects Week 4 scope only. KPIs are identified and justified, not yet calculated or visualised. Calculation, deeper analysis, and dashboard development are planned for later phases.
- Future dashboard work will depend on the dataset and Data Dictionary remaining stable. If HealthConnect provides an updated dataset or definitions, the data-quality assessment and analysis plan should be reviewed and, where necessary, re-run.